

Corrigendum: Super Ionic $\text{Li}_{3-2x}\text{M}_x(\text{OH})_{1-y}\text{N}_y\text{Cl}$ ($\text{M}=\text{Ca}$, W , $\text{N}=\text{F}$) Halide Hydroxide as an Anti- Perovskite Electrolyte for Solid-State Batteries [*J.* *Electrochem. Soc.*, **170**, 080512 (2023)]

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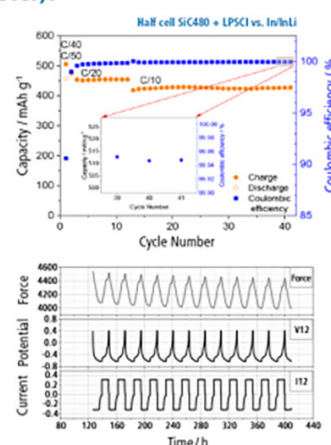
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This Corrigendum note and correction provides corrections to errors in the accepted manuscript posted online and in the published paper.

In the abstract, $\sigma(b)$ and $\sigma(g.b.)$ were mistakenly typed as $\sigma b = 23.7 \text{ mS cm}^{-1}$ and $\sigma gb = 9.2 \text{ mS cm}^{-1}$ at 140°C for $x = y = 0$. The sentence should be read as follows:

The impurity-free $\text{Li}_{3-2x}\text{M}_x(\text{OH})_{1-y}\text{N}_y\text{Cl}$ ($\text{M}=\text{Ca}, \text{W}, \text{N}=\text{F}$) are reported, having $\sigma_b = 9.6 \text{ mS cm}^{-1}$ and $\sigma_{gb} = 23.7 \text{ mS cm}^{-1}$ at 140°C for $x = y = 0$.

A new Fig. 3 with a corrected Fig. 3a is presented. The Nyquist curves of the AP sample at some temperatures, i.e., at temperatures above 90°C , were mistakenly drowned. These errors do not change any result or interpretation presented in the paper; just the figure is corrected. Figure 3 should appear as follows:

Also, Fig. 8 is modified to more accurately reflect its basis on Arrhenius plot of bulk conductivity in Fig. 3e. These modifications do not change any result or interpretation presented in the paper. Figure 8 should appear as follows.

In addition to the above corrections, a new Table SI containing all information about the samples and the results of the calculations is shown here, in which the geometrical dimensions of the samples and the

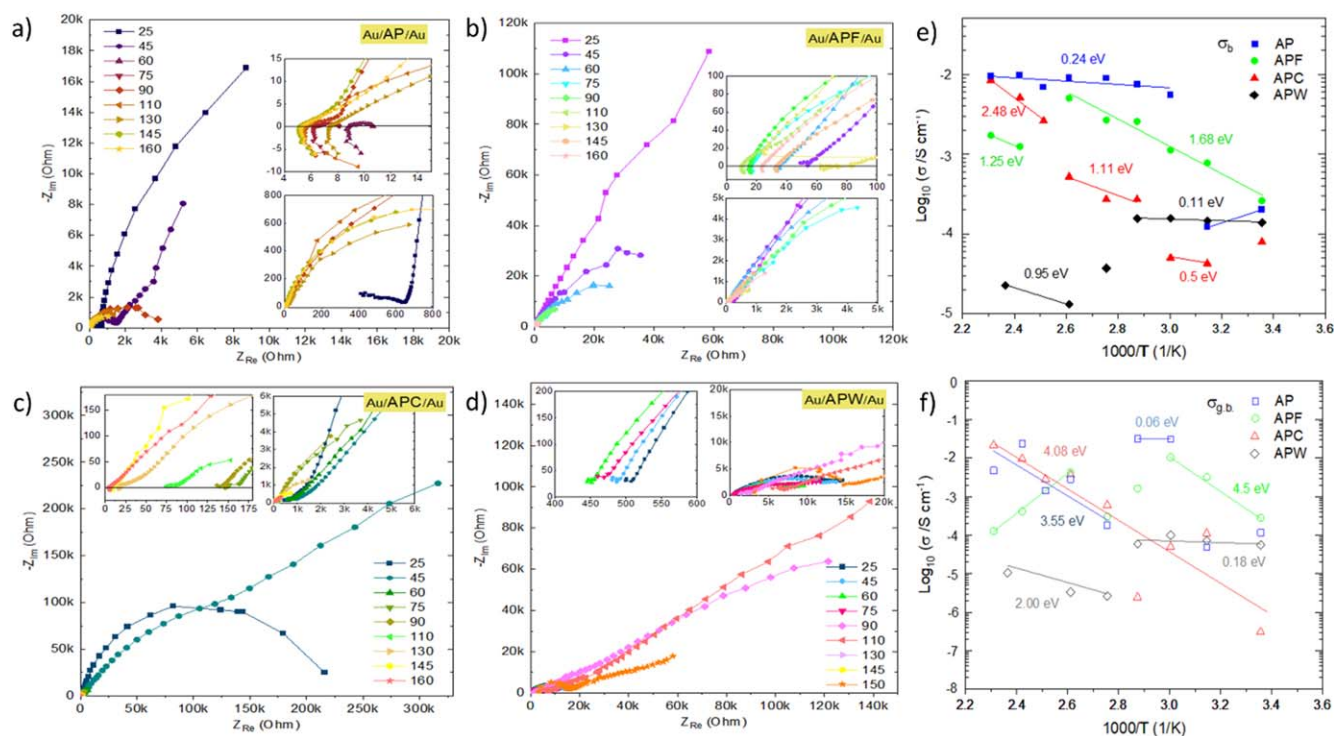


Figure 3. (Corrected). (a)–(d) Impedance spectroscopy Nyquist curves in a symmetric cell with the blocking gold electrode. Insets: high-frequency regions of the Nyquist curves. (e) Arrhenius plots of $\log(\sigma)$ vs $1/T$ with the measured activation energy for bulk ion conductivity. (f) Arrhenius plots of $\log(\sigma)$ vs $1/T$ with the measured activation energy for grain boundary ion conductivity. (g) Equivalent circuits for each sample.

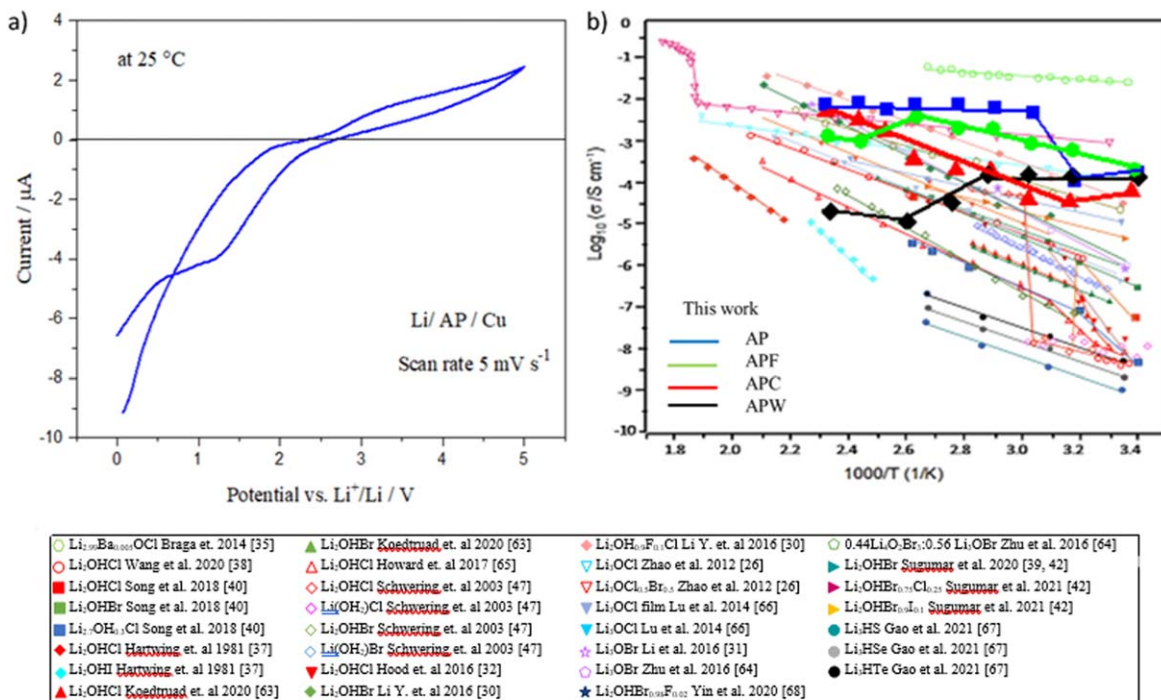


Figure 8. (Corrected). (a) cyclic voltammetry of the AP sample at RT. (b) Comparison of ion conductivity between the APs electrolytes in this work with the other solid anti-perovskite electrolytes, their chemical compositions are mentioned in the black border rectangular.

Table SI. Diameter, thickness, bulk resistance (R_b), conductivity (σ), and equivalent circuits for each sample at different temperatures derived from fitted EIS data.

Sample/temp. (°C)	Diameter (cm)	Thickness (cm)	R_b (Ω)	σ (S cm^{-1})	$\text{Log}_{10}\sigma$	Equivalent circuits
AP- 25	1.6	0.1	246	0.00020217	-3.69	
45			403	0.00012341	-3.90	
60			8.85	0.00561987	-2.25	
75			6.29	0.00790714	-2.10	
90			5.61	0.00886558	-2.05	
110			5.551	0.00895981	-2.04	
125			7.247	0.00686296	-2.16	
140			5.162	0.00963501	-2.01	
160			5.28	0.00941968	-2.02	
APC- 25	1.6	0.08	521	7.6369E-05	-4.11	
45			929.5	4.2806E-05	-4.36	
60			789.6	5.0391E-05	-4.29	
75			153.5	0.00025921	-3.58	
90			141	0.00028219	-3.54	
110			76.18	0.00052229	-3.28	

Table SI. (Continued).

Sample/temp. (°C)	Diameter (cm)	Thickness (cm)	R_b (Ω)	σ ($S\ cm^{-1}$)	Log_{10}^{σ}	Equivalent circuits
125	1.6	0.08	15.25	0.00260909	-2.58	
140			7.777	0.00511620	-2.29	
160			4.728	0.00841555	-2.07	
APF- 25			152	0.00026176	-3.58	
45			54.15	0.00073478	-3.13	
60			35.42	0.00112334	-2.94	
75			15.46	0.00257365	-2.58	
90			14.92	0.00266680	-2.57	
110			7.91	0.00503018	-2.29	
125			—	—	—	—
140	1.3	0.09	31.94	0.00124573	-2.90	
160			23.11	0.00172171	-2.76	
APW- 25			487	0.00013923	-3.85	
45			464	0.00014613	-3.83	
60			429.6	0.00015783	-3.80	
75			434	0.00015623	-3.80	
90			1818	3.7296E-05	-4.42	
110			5134	1.3207E-05	-4.87	
130			—	—	—	—
150			2984	2.2723E-05	-4.64	

equivalent circuits for each sample at different temperatures are presented. The coefficient of $\frac{L}{A}$ is very effective on the trend and on the ascending and descending slope of the points in the figure of Log_{10}^{σ} vs $1000/T$.

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